



5.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I understand how to use corresponding parts of congruent triangles in a proof.

I need help. I'm getting there, but I understand this
I can't get started. I need more practice. and I feel confident.



2. I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

I need help. I'm getting there, but I understand this
I can't get started. I need more practice. and I feel confident.



Unit 5, Lesson 8: The Proof Is in the Bag!



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.
- ☐ I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

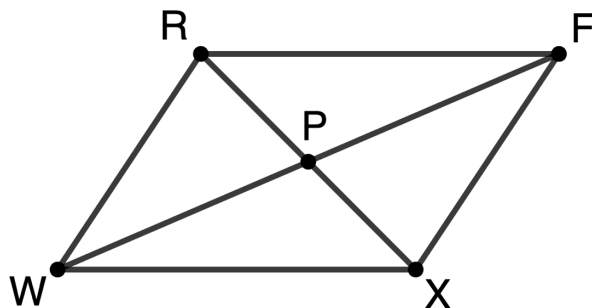


5.8.1 Warm-Up: Bell Work

Polygon $RFXW$ where diagonals $\overline{RX}$ and $\overline{FW}$ bisect each other at point P is shown.

Given: $\overline{RX}$ and $\overline{FW}$ bisect each other at point P .

Prove: $\triangle RPF \cong \triangle XPW$



1. Complete the two-column proof.

Statement	Reason
1. $\overline{RX}$ and $\overline{FW}$ bisect each other at point P	1. Given
2.	2. Definition of _____ bisector
3. $\overline{RP} \cong \underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}} \cong \overline{WP}$	3. Definition of _____
4.	4.
5. $\triangle RPF \cong \triangle XPW$	5.



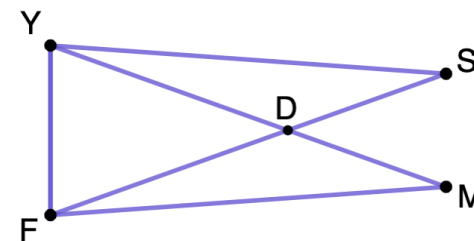
5.8.2 Activity: It's in the Bag!

With your group, complete each proof using the statement and reason cards provided.

Proof A

Given: $\angle YSD \cong \angle FMD$
 $\triangle YDF$ is isosceles with base $\overline{YF}$

Prove: $\overline{FM} \cong \overline{YS}$

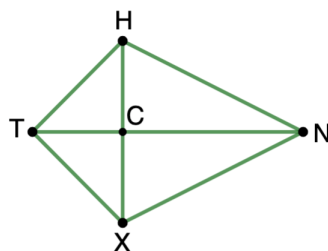


Statement	Reason

Proof B

Given: $\overline{TN}$ bisects HTX
 $\overline{HT} \cong \overline{XT}$

Prove: $\overline{TN}$ bisects $\angle HNX$

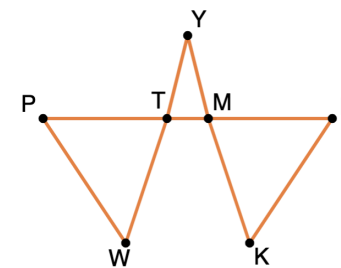


Statement	Reason

Proof C

Given: $\angle WPT \cong \angle KFM$
 $\overline{PW} \cong \overline{FK}$
 $\angle YMT \cong \angle YTM$

Prove: $\triangle PWT \cong \triangle FKM$

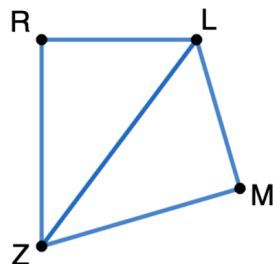


Statement	Reason

Proof D

Given: $\overline{LM} \cong \overline{LR}$
 $\overline{LR} \perp \overline{RZ}$
 $\overline{LM} \perp \overline{MZ}$

Prove: $\angle MZL \cong \angle RZL$

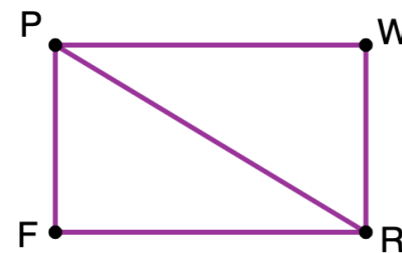


Statement	Reason

Proof E

Given: $\overline{PF} \cong \overline{WR}$
 $\overline{PW} \cong \overline{FR}$

Prove: $\overline{PF} \parallel \overline{WR}$

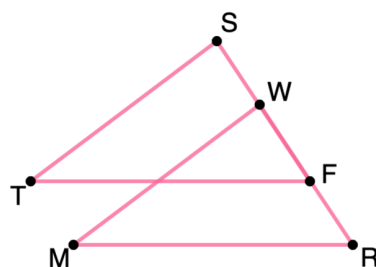


Statement	Reason

Proof F

Given: $\overline{RF} \cong \overline{SW}$
 $\angle TSF \cong \angle MWR$
 $\overline{FT} \parallel \overline{MR}$

Prove: $\triangle TSF \cong \triangle MWR$



Statement	Reason



5.8.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about writing proofs.